

Astrodynamics I: Midterm Exam

April 27, 2013 11:00AM-12:40PM (100 minutes)

1. (30) Two-Body Problem

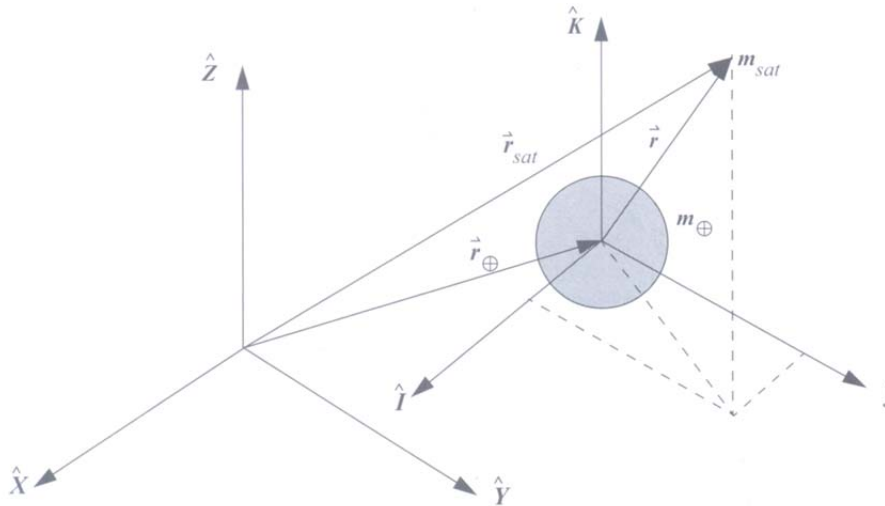


Figure 1-12. Geometry for Two Bodies in an Inertial Reference Frame. XYZ is assumed to be an inertial coordinate system. IJK is displaced from XYZ, but does not rotate or accelerate with respect to XYZ.

- (5) State precisely four underlying assumptions for deriving the two-body equations of motion.
- (10) Using the notations in Figure 1-12, derive the equations of motion of a satellite (sat) with respect to the primary body ($\oplus$) in the 2 body problem.
- (5) Show that the specific angular momentum is constant.
- (10) Show that the specific energy is constant.

2. (30) Kepler's Equation

- (20) Using the notations in Figure 2-2 and Kepler's Second and Third Laws, derive geometrically Kepler's equation in basic form:

$$\sqrt{\frac{a^3}{\mu}} = \frac{t - T}{E - e \sin E}$$

Hint: You may use the following relations without proof.

4. (20) Discuss the following terminologies **succinctly** and **precisely**.
 - a. (5) Flight Path Angle
 - b. (5) Kepler's Problem
 - c. (5) True Longitude of Periapsis for Elliptical Equatorial Orbit
 - d. (5) Argument of Latitude for Circular Inclined Orbit